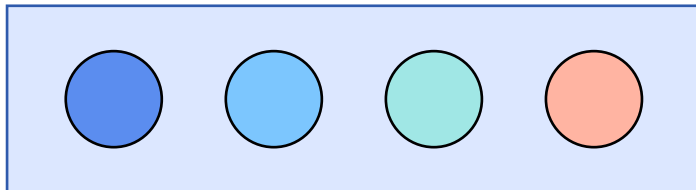


■ Multimodal Multiplayer Dyslexia Learning System

Creative User Manual

Interactive Educational Platform for Dyslexia Support

Gamified Learning • Multiplayer • Accessibility



1. Introduction

The Multimodal Multiplayer Dyslexia Learning System is an interactive educational platform developed to help children with dyslexia improve reading, cognitive, communication, and collaborative learning skills. The platform combines:

- A PC-based multiplayer learning game built using Unreal Engine
- A Flutter-based mobile application
- Real-time multiplayer synchronization
- Voice communication and adaptive learning systems

The system creates an engaging and stress-free environment where children can learn through interactive gameplay and multimodal feedback.

2. System Requirements

■ PC Application Requirements

- Windows 10 or higher
- Minimum 8GB RAM
- Stable internet connection
- Integrated or dedicated graphics support

■ Mobile Application Requirements

- Android 8 or newer
- Touchscreen support
- Internet/Wi-Fi connection

3. System Components

■ PC Game Environment

Interactive multiplayer virtual world containing educational activities and collaborative challenges.

■ Mobile Application

Supports tracing, drawing, touch interaction, and dyslexia-friendly input activities.

■ Voice Chat System

Enables safe real-time communication between players during collaborative activities.

■ Server Synchronization

Keeps all gameplay, scores, and learning progress synchronized in real time.

4. Getting Started

Step 1 — Launch the Game

1. Open the PC game application.
2. Log in or create a user profile.

Step 2 — Connect Mobile App

1. Open the mobile application.
2. Enter the connection code or join the same network session.

Step 3 — Join a Session

1. Select Single-player or Multiplayer mode.
2. Create or join a game session.

5. Gameplay & Interaction

■ PC Gameplay

- Navigate the 3D virtual environment
- Interact with learning objects
- Solve word-based challenges
- Collaborate with other players

📱 Mobile Interaction

- Letter tracing activities
- Color selection tasks
- Dot-connecting exercises
- Real-time input synchronization with PC game

■ Voice Communication

- Built-in voice chat system
- Mute/unmute options
- Safe anonymous communication environment

6. Key Features

■ Reward-Based Learning

Players earn rewards, badges, and achievements without stressful competitive leaderboards.

■ Adaptive Learning

The system automatically adjusts difficulty levels according to user performance and learning pace.

■ Multimodal Feedback

- Visual cues and highlighting
- Audio guidance and phoneme support
- Mobile vibration feedback

7. Game Modes

■ Single-Player Mode

- Independent practice
- Personal progress tracking
- Self-paced learning

■ Multiplayer Mode

- Real-time teamwork
- Shared educational challenges
- Communication and collaboration development

8. User Interface Overview

Main Menu

Access gameplay, settings, and user profiles.

Game Screen

Displays the 3D environment, objectives, and educational tasks.

Mobile Interface

Provides tracing tools, feedback prompts, and drawing activities.

Voice UI

Contains mute/unmute controls and communication indicators.

9. Data & Progress Tracking

The system tracks:

- Task completion

- Accuracy levels
- Number of attempts

- Engagement time

This data is used to: • Personalize learning pathways

- Adjust challenge difficulty
- Improve educational outcomes

10. Troubleshooting

Issue	Solution
Cannot connect mobile app	Check Wi-Fi or network connection
Voice chat not working	Enable microphone permissions
Game lagging	Reduce graphics settings
Sync delay	Ensure stable internet connection

11. Safety & Accessibility

- Dyslexia-friendly fonts and color palettes
- Reduced stress learning environment
- Safe communication system
- Non-competitive gameplay structure
- Accessibility-focused interactions

12. Future Updates

- Additional language support
- AI-based personalized learning recommendations
- Expanded game worlds and educational levels
- Enhanced accessibility tools

13. Contact & Support

■ **Email:** IT22610034@my.sliit.lk

■ **Institution:** SLIIT

Thank You!

This system was designed to provide an inclusive, engaging, and supportive learning experience for children with dyslexia through modern multiplayer technology and adaptive educational design.